

PII-TRACE: A Benchmark for Context-Aware PII Detection in Multi-Turn LLM Conversations

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Abstract

LLM assistants and agentic systems log long multi-turn conversations. AI providers often scan these conversations for Personally Identifiable Information (PII) and mask the PII before storing or processing conversation data. Yet most PII detectors and benchmarks target self-contained records rather than cross-turn evaluation. To evaluate PII detection across turns in multi-turn conversations, we introduce PII-TRACE (Tracing Recurring PII Across Conversational Exchanges), to our knowledge the first PII benchmark to assess whether detectors identify PII in conversational contexts and cover every mention of a recurring identifier across turns. PII-TRACE contains 13,148 synthetic multi-turn dialogues in 13 languages with character-level spans and identifier clusters. Across eleven baselines, including frontier LLMs, no detector achieves full entity-level coverage without substantial false positives on PII-free conversations, and single-pass reading loses a third of the gold characters on long dialogues. To close this gap, we introduce PII-Tracer, a compact 0.6B-parameter detector trained with conversation-level supervision. PII-Tracer attains the highest entity-level coverage of any system we evaluate and also performs strongly on standard single-record benchmarks. We will release both the benchmark and detector upon publication.

1 Introduction

Scanning text for Personally Identifiable Information (PII) is a standard component of data pipelines in modern LLM systems. Providers detect and remove PII when curating pretraining corpora (Soldaini et al., 2024; Laurençon et al., 2022; Li et al., 2023; Grattafiori et al., 2024) and before storing or reusing

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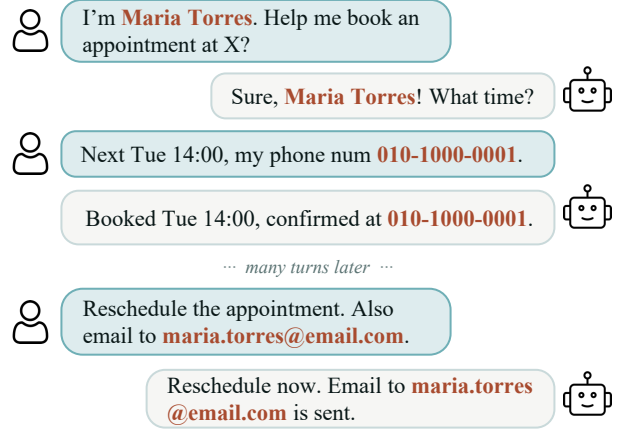


Figure 1: One identifier (highlighted) recurs across a conversation. PII-TRACE’s *consistent-detection* metric counts it as covered only if every mention is found.

collected data. This process helps satisfy data protection regulations (European Union, 2016) and reduces the risk that models memorize and later reveal sensitive information (Carlini et al., 2021). Increasingly, PII detectors are built on transformer-based language models, including a recently released open-weight bidirectional token classifier (OpenAI, 2026c).

Existing PII detectors are typically designed for relatively short, self-contained records rather than the long, multi-turn conversations common in LLM assistants and agentic systems. Within a single conversation, users may paste emails or medical reports, switch between languages, and interact with assistants that call external tools or share context with other agents (Yao et al., 2023; Xi et al., 2023; Packer et al., 2023). Detectors designed to process isolated records may therefore struggle with the length, complexity, and contextual dependencies of real-world conversations.

Conversational context determines what counts as PII and which mentions must be detected. The same text span can be PII in one conversation but not in another. For example, a name is PII when it refers to the user, but not when it refers to a public figure or a place named after that person. Moreover, the same identifier may recur throughout a conversation. In Figure 1, for example, the name *Maria Torres* appears across multiple turns. A detector must identify every occurrence of a recurring identifier, as missing even one leaves that occurrence unprotected. This requirement becomes especially important when conversations are passed to external tools or other agents, where private content may be further exposed (Wang et al., 2025; Yagoubi et al., 2026). PII detection in conversations is therefore inherently a **context-aware** task.

However, established PII benchmarks are primarily document- or record-oriented rather than conversational (ai4Privacy, 2024; Pilán et al., 2022; Stubbs et al., 2015). In our experiments, we identify three properties that these benchmarks do not adequately evaluate. The first is *cross-turn consistency*: every mention of a recurring identifier should be detected. The second is avoiding *long-context degradation*, where recall decreases as conversation length grows (Liu et al., 2024; OpenAI, 2026c). The third is the ability to handle *multilingual mixed-medium text*, where languages may switch within a dialogue and prose may be interleaved with code and structured records. As a result, strong performance on existing benchmarks does not necessarily translate into effective PII detection in real-world conversational settings. The closest concurrent efforts, such as REDACT (Vats et al., 2026) and RedactionBench (Brynjólfsson et al., 2026), cover more languages and longer documents, but do not provide explicit cross-turn mention chains needed to evaluate cross-turn consistency.

This paper asks: *Do PII detectors consistently cover recurring identifiers in multi-turn LLM conversations?*

To answer this question, we introduce PII-TRACE, to our knowledge the first PII benchmark with an explicit cross-turn detection task for multi-turn conversations. PII-TRACE comprises 13,148 synthetic dialogues derived from production assistant traffic, with character-level annotations spanning

nine identifier types. Mentions of the same identifier are further linked into entities, allowing us to evaluate whether a detector follows an identifier across the entire conversation. We call this property *consistent detection*: an entity is considered covered only if every mention of that entity is detected. Conversations in PII-TRACE range from fewer than 1,000 to more than 100,000 characters, enabling us to measure robustness to long-context degradation. The benchmark also includes conversations that switch languages and interleave prose with code, tables, and structured records, spanning 13 languages in total. It therefore enables evaluation of detectors on multilingual mixed-medium text. Finally, PII-TRACE includes PII-free conversations for measuring false positives.

We find that existing detectors struggle to consistently cover recurring identifiers across conversations. Across eleven baselines, ranging from open-source detectors to frontier LLMs prompted for PII detection, no public detector achieves strong consistent detection without flagging most PII-free conversations. Frontier LLMs are substantially more precise, but still miss many PII mentions.

To close this gap, we introduce PII-Tracer, a compact 0.6B-parameter detector trained on multilingual conversations. PII-Tracer labels an entire 4,096-token window in a single pass. Longer conversations can be processed using overlapping sliding windows, which substantially improves recall on long conversations (§4). PII-Tracer achieves the highest consistent detection and character-level F1 among all systems we evaluate, while using only a small fraction of the parameters of frontier LLMs (§3.7, §4).

Contributions.

- We introduce PII-TRACE, a multi-turn PII benchmark with an explicit cross-turn detection task. It includes long, multilingual, and mixed-medium conversations, with mentions linked into entities for evaluating *consistent detection* across turns.
- We evaluate eleven PII detectors, ranging from open-source specialized models to frontier LLMs. Public specialized detectors achieve high recall but low precision, while frontier LLMs are more precise but miss many mentions. For most detectors, recall also degrades substantially as conversations grow longer.

- We introduce PII-Tracer, a compact 0.6B-parameter PII detector trained on conversational data. It achieves better coverage of recurring identifiers than all other evaluated detectors, including much larger frontier LLMs.

2 Related Work

PII benchmarks. Existing benchmarks cover synthetic records, including ai4privacy, Nemotron-PII, and SPY (ai4Privacy, 2024; NVIDIA, 2025b; Savkin et al., 2025); legal documents with within-document coreference, such as TAB (Pilán et al., 2022); longitudinal but non-conversational clinical records, such as i2b2/UTHealth (Stubbs et al., 2015); and newswire named-entity recognition (NER), such as CoNLL-2003 (Tjong Kim Sang and De Meulder, 2003). Concurrent work includes REDACT, which covers mixed formats and multi-turn chats without cross-turn mention chains (Vats et al., 2026); the document-oriented RedactionBench (Brynjólfsson et al., 2026); and the query-focused PII-Bench and CAPID (Shen et al., 2026; Ponomarenko et al., 2026). None provides explicit mention chains across turns of a longitudinal user–assistant conversation (Table 1).

PII detectors. Existing detectors range from rule- and NER-based systems, such as Microsoft Presidio (Microsoft, 2024), to learned span models, including the OpenAI Privacy Filter (OpenAI, 2026c), GLiNER2-PII (Zaratiana et al., 2026), and open-source detectors from the Piirinha (iiiorg, 2024) and OpenMed (OpenMed Science, 2026) series. We evaluate seven of these detectors on conversations in §4.

Contextual and agentic privacy. Privacy is contextual: whether an information flow is appropriate depends on the context in which it occurs (Nissenbaum, 2004). LLMs can leak private information in inappropriate contexts (Miresghallah et al., 2024), infer hidden attributes from contextual cues (Staab et al., 2024), and reproduce memorized training data (Carlini et al., 2021; Kim et al., 2023). Agentic privacy benchmarks test whether agent actions respect privacy norms (Shao et al., 2024) and measure information leakage through tool use and inter-agent interactions (Wang et al., 2025; Yagoubi et al., 2026). Separately, long-context studies show that model performance degrades as input length increases (Liu

et al., 2024; Hsieh et al., 2024; Bai et al., 2024). PII-TRACE is complementary to this work: it evaluates span-level PII detection across an entire conversation.

3 The PII-TRACE Benchmark

3.1 Problem formulation

We study PII detection in long, multi-turn conversations. Given a complete user–assistant dialogue, the task is to identify every text span that refers to a private individual. We adopt an operational definition of PII as information that identifies a specific person, either on its own or when combined with other information. This definition draws on the GDPR (European Union, 2016), the ISO/IEC 29100 privacy framework (ISO/IEC, 2011), U.S. federal guidance (McCallister et al., 2010), and scholarship on contextual identifiability (Schwartz and Solove, 2011).

Consider the name *Maria Torres* in Figure 1. Whether this span constitutes PII cannot be determined from the string alone. If a user states their name, the span reveals personal information. The same string does not identify anyone when the assistant introduces it as a fictional placeholder in an example. Distinguishing among these cases requires reasoning about the surrounding conversational context, making PII detection a **context-aware** task.

Formally, a conversation is a sequence of m user and assistant turns concatenated into one text,

$$x = \tau_1 \parallel \tau_2 \parallel \cdots \parallel \tau_m \quad (1)$$

where turn τ_i occupies the character positions T_i of x . A detector is graded on all of x . It may read the text in one window or several (a choice we leave to its inference policy), but it must commit to a single set P of character positions marked as PII. We do not require it to say which marked positions belong to the same entity. The task is detection only.

The nine identifier types shown in Table 2 cover the label sets of deployed detectors (OpenAI, 2026c; Zaratiana et al., 2026). Let G be the set of gold PII characters. An *entity* $E \subseteq G$ is the set of characters belonging to a single identifier, with repeated mentions grouped by coreference, and the entities $\mathcal{E}$ partition G . We call an entity *cross-turn* if its characters appear in more than one turn in the conversation,

$$|\{i : E \cap T_i \neq \emptyset\}| \geq 2. \quad (2)$$

Table 1: PII-TRACE vs. representative PII benchmarks: only PII-TRACE pairs multi-turn conversational text with cross-turn identifier clusters (TAB annotates coreference within single documents; REDACT includes flat chats among mixed formats but defines no cross-turn task). Sizes approximate.

Benchmark	#Ex.	Langs	Setting	Entity-cluster	Cross-turn
ai4privacy (ai4Privacy, 2024)	225k	6	short document	✗	✗
Nemotron-PII (NVIDIA, 2025b)	100k	1	document	✗	✗
SPY (Savkin et al., 2025)	8.7k	1	QA (author)	✗	✗
TAB (Pilán et al., 2022)	1.3k	1	legal doc	✓	✗
REDACT (Vats et al., 2026)	13.4k	25	mixed incl. chat	✗	✗
PII-Bench (Shen et al., 2026)	2.8k	1	query + description	✗	✗
RedactionBench (Brynjólfsson et al., 2026)	0.2k	1	document	✗	✗
PII-TRACE (ours)	13,148	13	multi-turn conv.	✓	✓

Table 2: The nine identifier types, with gold mention counts over all three splits (37,431 total; every occurrence counts separately).

Type	Covers	Mentions
private_person	name of a private person: full name, name part, handle	22,973
private_date	date identifying a person	3,381
private_url	personal URL or IP address	2,657
private_address	postal address, precise location	2,521
account_number	national ID, SSN, IBAN, card, account ids	2,013
private_email	personal email address	1,626
private_phone	personal phone or fax	985
other_pii	residual annotator-marked identifiers	860
secret	credentials: passwords, keys, PINs	415

Achieving protection requires **consistent detection** because for recurring identifiers, missing even one occurrence leaves the identifier exposed. First, at the character level, a label is considered correct if a labeled position belongs to G . This measures how many PII texts are detected. Second, at the entity level, an entity is considered consistently detected if all characters of entity E are labeled ($E \subseteq P$). The evaluation setting (§4.1) embodies these two criteria into the precise metrics reported in this paper.

3.2 Design desiderata

Our design goals follow the actual deployment scenario: for the LLM assistant, the PII detector deals with dialogue logs, not the short, well-structured records common in existing corpora.

Splitting a dialogue into individual records results in the loss of significant structural information, since an identifier may reappear after many rounds. We

assign a cluster ID to each occurrence, enabling evaluation to track the identifier across the entire dialogue rather than a single location. Dialogue lengths vary widely, from a few hundred to over one hundred thousand characters, allowing us to measure recall based on input length. Dialogues are also not always monolingual natural language text: they may switch languages mid-conversation, and plain text may contain code, tables, or structured records. In our experiments, existing baseline detectors performed poorly primarily on dialogues with these characteristics (§4).

Pipeline. We synthesize the benchmark from real traffic in five steps (Figure 2), labeling and anonymization (steps 1, 2, §3.3), synthesis (steps 3, 4, §3.4), and the verification gates that decide whether a record (step 5) is released (§3.5).

3.3 Labeling and anonymization

We begin with large-scale real-world user-assistant dialogue samples collected from a production environment, transforming each dialogue into a de-identified template. Multiple state-of-the-art LLMs are used to annotate identifiers according to the nine types of annotation in Table 2 (Figure 2, step 1). Subsequently, recurring identifiers of the same type are grouped into the same entity through rule-based processing, and sensitive attribute layers are annotated in a separate process. In step 2, we remove each identifier and replace it with a placeholder. The template retains the multi-turn structure of the dialogue, as well as the position, type, and entity ID of each occurrence, while not retaining any of the original annotated values.

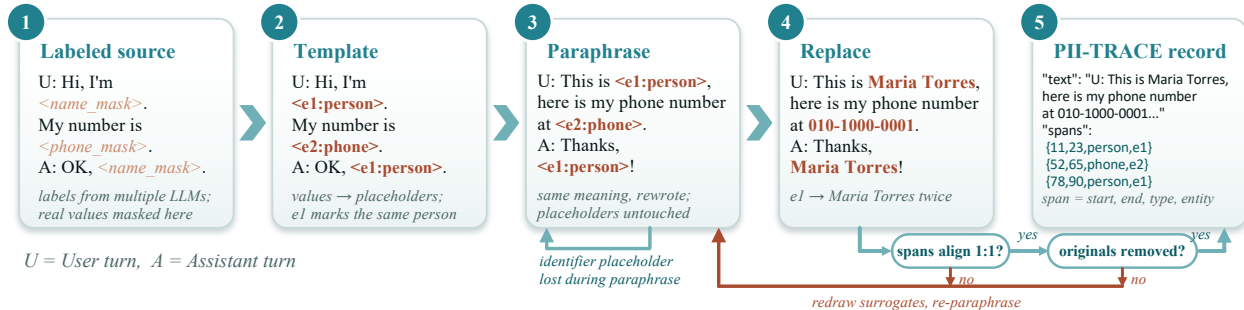


Figure 2: Constructing PII-TRACE, on one example (source values withheld): a labeled conversation becomes a typed, cluster-preserving template; each turn is paraphrased with placeholders intact, and each placeholder is replaced by its entity’s surrogate with exact offsets. Failed paraphrases are retried, and gate-flagged documents can re-enter synthesis with fresh surrogates before a final drop (§3.5). Steps 1 and 2 are the anonymization of §3.3, steps 3 and 4 the synthesis of §3.4, and the gates, guarding the released record (step 5), the verification of §3.5.

3.4 Synthesis

Each template in the anonymization phase generates a synthesized dialogue (Figure 2, steps 3 and 4). Surrogates are generated for each entity, matching their type, format, and geographic location, and using values that are formatted correctly but not usable. The same surrogate is used for every occurrence, ensuring identifier consistency within a document, while surrogates in different documents are generated independently. The language model then rewrites each round of dialogue (step 3) and inserts the surrogates (step 4), so both the textual representation and the annotated identifier values in the final benchmark are synthesized content. If a placeholder is lost during paraphrasing, the rewrite is regenerated up to three times (back loop in step 3); if it still fails, the document is flagged and proceeds to the verification checks in §3.5. This replacement strategy follows the clinical de-identification practice (Carrell et al., 2013; Stubbs et al., 2015).

3.5 Alignment, verification, and audit

Every tagged identifier in the benchmark is a surrogate that the pipeline placed (Figure 2, step 5), so the gold span can be accurately determined without manual annotation. The alignment step further ensures accurate positioning: after rewriting, we recalculate each surrogate’s character offset and append its type and entity ID.

Subsequently, the validation process verifies the correct construction of each document through three automated checks. First, the replacement value span

must correspond one-to-one with the gold mention, and all mentions of the same entity must share the same value. Second, no original values should be detected during independent rescanning using Microsoft Presidio and strict regular expressions. Third, each stored character offset must accurately extract the corresponding surrogate substring. The first two checks correspond to the decision nodes in Figure 2. A document that fails any gate is resynthesized with fresh surrogates and re-paraphrased turns (the loop back into step 3), and discarded if it still fails. Since these checks are rule-based, we also used a second, independent language model to audit a portion of the released documents as a fuzzy check; anything it flagged was manually reviewed against the document’s surrogate list. See Appendix E for the prompts used in the pipeline.

3.6 Composition

PII-TRACE contains 13,148 conversations, split into training, validation, and test sets with no source conversations shared between the different data partitions; 5,645 dialogues contain gold PII, and 7,503 are PII-free. Dialogue lengths range from less than 1,000 characters to over 100,000 characters, and thirteen languages each contain at least 100 dialogues. In dialogues containing PII, 63.8% of the dialogues contain entities mentioned multiple times, and 28.7% of the dialogues have entities that appear repeatedly across rounds; therefore, detecting only one mention often fails to cover the entire entity.

3.7 PII-Tracer: a context-aware detector

PII-Tracer models PII detection in the dialogue as token classification. One encoder pass reads a window (§3.1) of the dialogue x and generates contextual states $h_{1:T}$ for T tokens within it. Therefore, the annotation for each token is based on the surrounding dialogue window as context, rather than using the autoregressive prompting used in existing LLM baselines.

Architecture. PII-Tracer is a bidirectional encoder with 0.6B parameters, using a Qwen3 (Qwen Team, 2025) backbone adapted with masked-diffusion pre-training, reading a maximum of 4096 tokens per window. On the shared token states, the model uses a tagging head covering 37 BIOES labels (including a background tag O and $\{B, I, E, S\}$ for each of the nine types), and an auxiliary sensitivity head used during training to predicts whether the conversation contains sensitive content as an auxiliary training signal. Gold spans are mapped to tokenization and represented as BIOES tags; therefore, detection is modeled as per-token classification with the surrounding conversation window as context.

Training and decoding. We minimize

$$\mathcal{L} = \lambda_{\text{tok}} \mathcal{L}_{\text{tag}} + \lambda_{\text{sens}} \mathcal{L}_{\text{sens}}, \quad (3)$$

where $\mathcal{L}_{\text{tag}}$ is the class-weighted cross-entropy for BIOES tags, and $\mathcal{L}_{\text{sens}}$ is the binary cross-entropy for sensitivity logit, with weights of $\lambda_{\text{tok}}=1.5$ and $\lambda_{\text{sens}}=0.3$, respectively. We trained on AdamW for 3 epochs with 714k samples, including multilingual assistant conversations annotated by multiple frontier language models and samples from the ai4privacy corpus. Each training sample concatenates all turns in a conversation into plain text and annotates each mention on the conversation-level offset, enabling the model to judge each span in conjunction with the conversational context. Therefore, the same string can be supervised as a PII in one conversation and as background in another. During inference, per-token scores are decoded into typed spans while ensuring the tag sequences are valid; two learnable boundary biases adjust the tradeoff between precision and recall without retraining; for conversations longer than the window, decoding can be performed window-by-window (§4). More hyperparameter settings in Appendix C.

4 Experiments

4.1 Evaluation setup

Our evaluation covers twelve detectors: Microsoft Presidio (Microsoft, 2024), six learned span models (the OpenAI Privacy Filter with 1.5B parameters (OpenAI, 2026c), GLiNER2-PII (Zaratiana et al., 2026), zero-shot GLiNER-PII (NVIDIA, 2025a), Pii-irinha (iirg, 2024), and two OpenMed clinical PII models (OpenMed Science, 2026)), four frontier LLMs prompted zero-shot with the nine-type taxonomy (GPT-5.4 (OpenAI, 2026a), GPT-5.6-sol (OpenAI, 2026b), Claude Opus 4.8 (Anthropic, 2026a), and Claude Sonnet 5 (Anthropic, 2026b)), and ours PII-Tracer. We evaluate on the test split (1,922 documents) and report metrics as following:

- **Character P / R / F1:** A predicted character is considered a true positive if it lies within a gold span. Character F1 is our primary detection metric, measuring how many PII texts are successfully masked without considering span boundaries.
- **Span-Overlap and Span-Containment P / R / F1:** For overlap, a prediction is considered correct if it overlaps with any gold span. For containment, precision counts predicted spans contained within a gold span, while recall counts gold spans contained within a predicted span.
- **Consistent detection:** The percentage of gold entities whose mentions are covered. CD-multi is calculated only for multi-mention entities, while cross-turn CD is calculated only for entities that repeat across rounds.
- **Has-PII accuracy and FP₀:** The former is the document-by-document binary has-PII accuracy, and the latter is the false-positive rate on documents without PII.

4.2 Main results

Table 3 divides the twelve systems into two groups. Publicly available specialized baselines achieve high recall through over-marking. OpenMed models can cover more than 0.9 of gold characters, but at the cost of marking more than six times the gold character volume, resulting in character precision below 0.15. Presidio marks 16 times the gold volume, with a precision of only 0.045, primarily covering structured

Table 3: PII detection on the PII-TRACE test set (1,922 documents); all metrics label-agnostic, higher is better. The four frontier LLMs are prompted zero-shot; span scores match gold spans by overlap and by containment.

Detector	Character			Span-Overlap			Span-Containment		
	P	R	F1	P	R	F1	P	R	F1
Presidio (rule/NER) (Microsoft, 2024)	0.045	0.762	0.086	0.042	0.844	0.079	0.036	0.741	0.069
OpenAI Privacy Filter (1.5B) (OpenAI, 2026c)	0.358	0.785	0.492	0.266	0.817	0.401	0.380	0.363	0.371
GLiNER2-PII (Zaratiana et al., 2026)	0.211	0.475	0.292	0.181	0.525	0.269	0.173	0.514	0.259
GLiNER2-PII (zero-shot) (NVIDIA, 2025a)	0.261	0.785	0.392	0.209	0.890	0.338	0.272	0.483	0.348
Piiranhä v1 (iiiorg, 2024)	0.161	0.299	0.209	0.117	0.339	0.174	0.078	0.157	0.104
OpenMed PII (44M) (OpenMed Science, 2026)	0.136	0.901	0.236	0.094	0.942	0.170	0.040	0.786	0.077
OpenMed PII (434M) (OpenMed Science, 2026)	0.145	0.919	0.251	0.109	0.957	0.195	0.047	0.787	0.089
GPT-5.4 (OpenAI, 2026a)	0.466	0.569	0.512	0.482	0.547	0.512	0.434	0.547	0.484
GPT-5.6-sol (OpenAI, 2026b)	0.555	0.679	0.611	0.589	0.681	0.632	0.558	0.679	0.612
Claude Opus 4.8 (Anthropic, 2026a)	0.502	0.556	0.528	0.488	0.519	0.503	0.470	0.515	0.491
Claude Sonnet 5 (Anthropic, 2026b)	0.536	0.602	0.567	0.579	0.566	0.572	0.510	0.570	0.539
PII-Tracer (0.6B)	0.507	0.830	0.629	0.496	0.832	0.621	0.445	0.831	0.580

Table 4: Consistent detection by an entity’s mention count and, for multi-mention entities, by turn span: each cell is the fraction of that group’s entities with *every* mention covered. Best per row in **bold**.

		Presidio	OpenAI PF	GLiNER2-PII	GPT		Claude		PII-Tracer
					5.4	5.6-sol	Opus 4.8	Sonnet 5	
count	1	0.655	0.474	0.641	0.829	0.788	0.840	0.829	0.917
	2	0.671	0.397	0.356	0.413	0.689	0.333	0.449	0.873
	3–5	0.630	0.263	0.258	0.216	0.504	0.216	0.252	0.796
	6–10	0.627	0.182	0.073	0.082	0.464	0.045	0.118	0.691
turn	cross-turn	0.648	0.296	0.292	0.275	0.551	0.222	0.305	0.776
	within-turn	0.615	0.343	0.154	0.302	0.663	0.320	0.355	0.876

identifiers such as email and phone numbers, with less coverage of free text. Frontier general-purpose LLMs exhibit the opposite pattern. They have the highest character precision, approaching 0.5, while all publicly available specialized baselines do not exceed 0.36; however, they only cover gold characters from 0.56 to 0.68.

PII-Tracer combines the advantages of both. It can cover 0.830 gold characters, comparable to overmarking detectors, while achieving a precision of 0.507, comparable to frontier LLMs; its char-F1 is 0.629, the highest of all systems, making it the most balanced among the twelve detectors. The much larger parameter-scale GPT-5.6-sol only slightly surpasses it in span-level F1 (overlap 0.632 vs. 0.621, containment 0.612 vs. 0.580).

4.3 Cross-turn consistency

Covering characters is easier than covering every mention of an entity, since missing any mention leaves the entity exposed; multi-mention consistent

detection is therefore lower than character recall for all detectors. PII-Tracer covers 0.830 of gold characters, with a multi-mention consistent detection (CD-multi) of 0.794, the highest of all systems, significantly higher than GPT-5.6-sol (0.570) and other frontier LLMs (0.24–0.31). Aggregate CD, CD-multi, cross-turn CD, and PII-free false-positive rates for the twelve detectors are shown in Appendix D (Table 7), while Table 4 analyzes consistent detection by mention count (899 single-mention and 959 multi-mention entities, 790 of the latter appearing repeatedly across rounds). As the number of mentions increases, consistent detection declines sharply for learned detectors (GLiNER2-PII from 0.641 for a single mention to 0.073 for 6–10 mentions) and frontier LLMs (GPT-5.6-sol from 0.788 to 0.464, Claude Opus 4.8 from 0.840 to 0.045), whereas PII-Tracer declines more gradually (from 0.917 to 0.691) and performs best across all groups, including cross-turn entities (0.776). Presidio consistently performs at

Table 5: PII-Tracer character P/R/F1 by conversation length, read in a single 4,096-token window. Docs: test documents per bucket; PII docs: those with gold spans; characters pooled per bucket.

Length	Docs	PII docs	P	R	F1
<1k	167	49	0.392	0.975	0.559
1k–10k	1,292	505	0.516	0.955	0.670
≥10k	463	243	0.507	0.687	0.583
Total	1,922	797	0.507	0.830	0.629

around 0.63 across all groups because it labels most structured values rather than selectively covering a particular entity.

4.4 Document-level flagging

In practice, Guardrail must also determine whether a conversation contains PII, as incorrectly labeling clean conversations incurs costs. Has-PII accuracy reveals over-flagging issues: Presidio and OpenMed models score between 0.43 and 0.45, below the Has-PII baseline because they label PII in almost every conversation; frontier LLMs score between 0.76 and 0.83; and PII-Tracer reaches 0.759, the highest among specialized detectors and close to frontier LLMs. The complementary metric FP_0 is shown in Table 7: all public specialized detectors label more than half of the PII-free documents, while the proportion for PII-Tracer is 0.385. This reflects the cost of over-marking at the document level, as predicted spans scattered throughout clean conversations become false flags.

4.5 Single-window degradation

As dialogue length increases, recall decreases. In a single 4096-token window, PII-Tracer covers 0.975 of gold characters in dialogues shorter than 1k characters, 0.955 in dialogues between 1 and 10k characters, and drops to 0.687 for dialogues longer than 10k characters (Table 5). Precision does not decrease with length; it is lowest in the shortest range (0.392), and around 0.51 in other ranges. This is primarily due to single-window truncation: using 50%-overlap sliding windows decoding for the same checkpoint can improve character recall from 0.83 to 0.97 and multi-mention CD from 0.79 to 0.95 (Appendix D). Longer dialogues contain a larger absolute number of identifiers, so documents with decreased recall

	en	de	fr	it	ru	ko
Presidio	.178	.049	.133	.028	.033	.119
OpenAI PF	.556	.605	.517	.283	.275	.358
GLiNER2-PII	.300	.309	.356	.159	.280	.266
GLiNER-PII	.379	.467	.470	.247	.294	.460
Piiranhä	.196	.254	.229	.178	.234	.187
OpenMed 44M	.213	.360	.306	.161	.216	.189
OpenMed 434M	.225	.341	.299	.150	.246	.284
GPT-5.4	.511	.522	.528	.466	.485	.610
GPT-5.6-sol	.639	.661	.561	.496	.503	.652
Claude Opus 4.8	.542	.519	.500	.486	.578	.489
Claude Sonnet 5	.548	.528	.602	.545	.645	.538
PII-Tracer	.623	.735	.633	.676	.651	.616

Figure 3: Character F1 by language for six test languages covering the Latin, Cyrillic, and Hangul scripts.

face a greater risk of missed detections. The sliding-window method incurs a slight loss of precision, but the two strategies can be chosen at decode time, so a suitable operating point can be selected during deployment without retraining.

4.6 Multilingual mixed-medium text

Figure 3 shows the character F1 scores for six language subsets in the corpus, covering Latin, Cyrillic, and Hangul scripts; full language-specific results will be available upon release. Each language was evaluated on all test documents, ranging from 928 for English to 61 for Italian. System rankings varied across different scripts. Presidio covered 0.79 of gold characters on English but only 0.42 on Russian; OpenMed 434M’s recall dropped from 0.94 for English to 0.69 for Korean; OpenAI Privacy Filter maintained recall across different scripts, thus its F1 score dropped to 0.28 on Italian and Russian, primarily due to precision. PII-Tracer achieved the highest character F1 score in four of the six languages (all between 0.616 and 0.735; GPT-5.6-sol scored slightly higher in ‘en’ and ‘ko’), and the highest consistent detection across all six languages, ranging from 0.80 to 0.93. Because rankings vary across different scripts, selecting a detector solely based on overall or English scores may not suit for real-world language combinations.

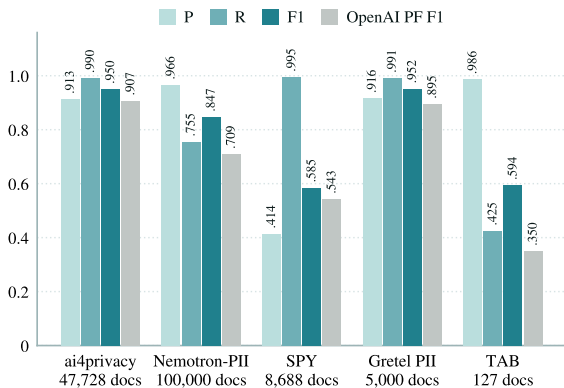


Figure 4: PII-Tracer on five external PII benchmarks (label-agnostic character P/R/F1); gray: the OpenAI Privacy Filter under identical evaluation.

4.7 External benchmarks

Figure 4 shows the results of PII-Tracer on five external PII benchmarks, including two public PII datasets used in the OpenAI Privacy Filter model card: ai4privacy and SPY. PII-Tracer outperforms the Privacy Filter on all benchmarks: char F1 0.950 vs 0.907 on ai4privacy (ai4Privacy, 2024), 0.847 vs 0.709 on Nemotron-PII (NVIDIA, 2025b), 0.585 vs 0.543 on SPY (Savkin et al., 2025), 0.952 vs 0.895 on the Gretel PII masking set (AI, 2024), and 0.594 vs 0.350 on TAB (Pilán et al., 2022). TAB is the only benchmark that includes real human-annotated text, and at the same precision, PII-Tracer achieves twice the recall. The improvement does not stem from a shared synthesis style. Table 3 indicates that single-record detectors perform poorly in dialogue scenarios, whereas PII-Tracer is competitive in both settings. Since its training mixture includes single-record data, we do not attribute this result entirely to conversational supervision.

5 Conclusion

In this work, we propose PII-TRACE, a benchmark for PII detection in multi-turn LLM conversations, and PII-Tracer, a compact 0.6B detector trained on such data. Experiments show that existing detectors either over-label PII-free conversations or miss some mentions of recurring identifiers; in contrast, PII-Tracer more completely covers recurring identifiers and performs well on standard external benchmarks. We call for greater attention to PII detection in multi-

turn LLM conversations.

Limitations

PII-TRACE is designed as a controlled benchmark of PII detection in multi-turn user–assistant conversations. Its conversations are synthetic reconstructions derived from the structure of production assistant traffic. This construction supports public release with exact span offsets and consistent mention chains, but it does not reproduce the source distribution verbatim. The reported results should therefore be read as measurements of conversational PII detection rather than estimates for any particular production workload.

The current release focuses on conversation text. Tool calls, inter-agent messages, and multimodal inputs fall outside its scope, although they are natural extensions for studying how personal data moves through agentic systems. Evaluation on conversations from additional assistants and domains would also provide a broader test of transfer beyond the setting represented here.

Finally, each baseline is evaluated under a single inference configuration. Alternative prompts, thresholds, context-window policies, or future model updates may change absolute scores. The benchmark is thus best suited to comparing detector behavior within the evaluation setup used here, rather than certifying production safety or regulatory compliance.

Ethical Considerations

PII-TRACE is built to strengthen protective systems: the intended use is evaluating and improving PII detectors, not re-identifying anyone. The source conversations were accessed and processed under the originating provider’s terms of service and privacy policy; access was limited to authorized project members under the provider’s data-access controls, and the manual review of audit-flagged documents took place under the same controls. No crowdworkers or external annotators were involved: identifiers were labeled by prompted language models, and human review was limited to the project team. We release only synthetic data: no source conversation is published, every tagged identifier is a fabricated, leak-checked surrogate, and the known residual, untagged personal names in some non-English text, is documented in the datasheet (Appendix A). The benchmark and the de-

tector will be released under the MIT license. Scores on synthetic data are proxies for, not guarantees of, behavior on real traffic, and the benchmark certifies neither production safety nor regulatory compliance. Language models assisted with writing and engineering in this project; the authors reviewed and verified all content and results.

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A Datasheet for Datasets

Following Geburu et al. (2021), we answer the core questions inline; the release will include a data card with the full questionnaire.

Motivation. PII-TRACE was created to evaluate PII detection in *multi-turn conversational context*, an axis missing from prior sentence- or record-level resources.

Composition. Each instance is a synthetic multi-turn user–assistant conversation with character-level identifier mentions over nine types, per-mention cluster ids, and a separate sensitive-attribute layer. The corpus has 13,148 conversations across 13 languages (train/validation/test = 9,202 / 2,024 / 1,922), with per-record metadata. Figure 5 shows a complete released record.

Collection and annotation. The corpus is derived from a deduplicated sample of real user–assistant conversations from a production assistant; a “turn” is one delivered message. To anonymize each conversation, multiple frontier LLMs locate the identifiers,

following a written guideline, and a rule-based pass groups repeated mentions of the same typed identifier into an entity; sensitive attributes are marked separately. We inherit these labels as PII-TRACE’s gold. The labeler prompt and guideline are summarized in Appendix E, and the release will include them.

B Full Taxonomy and Crosswalk

Table 6 lists the nine identifier types with their subtypes and definitions, plus a crosswalk to standard regulatory categories; the full row-by-row mapping to OMB, NIST SP 800-122, ISO/IEC 29100, and HIPAA will accompany the release. The label space subsumes those of the deployed detectors we evaluate, so their outputs map directly into our types; the residual `other_pii` covers identifiers outside the eight concrete types. The eight sensitive-attribute classes (health condition, religion, race/ethnicity, sexuality, political view, disability, age, gender) form the separate attribute layer and are not part of the identifier taxonomy.

C Detector: Additional Details

Training details. We train with AdamW (learning rate 10^{-4} , effective batch 256) on 8 data-parallel GPUs for 3 epochs, selecting the best checkpoint on held-out loss. The training data and PII-TRACE’s sources share the same production traffic and matching conversation ids.

Frontier-LLM detector protocol. The four frontier LLMs in Table 3 are prompted zero-shot, with no few-shot examples. A system message states the task and the nine identifier types and asks for a JSON list of `{text, type}` objects whose `text` is copied verbatim from the input (Appendix E). Documents are read in 8,000-character windows with 400-character overlap; each returned string is mapped back to character offsets by exact substring match, and offsets are merged and de-duplicated across windows.

D Extended Results

Full detection results. Table 7 completes the entity-level view of §4.3 for all twelve detectors. PII-Tracer leads on all three coverage columns (CD 0.853, CD-m 0.794, xCD 0.776). Its FP_0 of 0.385 is comparable to the frontier LLMs (0.231–0.386) and

```
{
  "id": "conv_00042",
  "language": "en",
  "length_bucket": "<1k",
  "document_format": "unstructured",
  "clean_negative": false,
  "text": "U: I'm Dana Okoye, my number is +00 10 1111 0000.\nA: Thanks, Dana Okoye, I've noted the callback line.\nU: Please keep Dana Okoye on file.",
  "spans": [
    {
      "start": 7,
      "end": 17,
      "label": "private_person",
      "subtype": "full_name",
      "entity_id": 1,
      "entity_mentions": 3
    },
    {
      "start": 32,
      "end": 48,
      "label": "private_phone",
      "subtype": "phone",
      "entity_id": 2,
      "entity_mentions": 1
    },
    {
      "start": 61,
      "end": 71,
      "label": "private_person",
      "subtype": "full_name",
      "entity_id": 1,
      "entity_mentions": 3
    },
    {
      "start": 118,
      "end": 128,
      "label": "private_person",
      "subtype": "full_name",
      "entity_id": 1,
      "entity_mentions": 3
    }
  ],
  "attribute_spans": []
}
```

Figure 5: A complete released record (values fabricated). Offsets index the text field; `entity_id` links repeated mentions of one identifier, and `entity_mentions` is the cluster size used by consistent detection.

Table 6: The nine PII-TRACE identifier types: sub-types, one-line definition (a span is labeled only when it identifies a *private* person; placeholders and public entities are excluded), and regulatory crosswalk.

Type	Sub-types	Definition / crosswalk
private_person	full name, name part, social handle/username	name of a private person; OMB/NIST direct, HIPAA name
private_email	email address	personal email; NIST, HIPAA electronic mail
private_phone	phone/fax number	phone of a private person; NIST, HIPAA
private_address	postal address, precise geo	location of a private person; HIPAA geographic
private_url	profile URL, IP address	URL/IP for a private audience; NIST online identifier
private_date	date of birth, identifying date	datetime identifying a person; HIPAA dates
account_number	national ID, SSN, IBAN, credit card, sort code, id number	account/record identifier; NIST, HIPAA account no.
secret	password/API key, CVV/PIN	credential; NIST authentication data
other_pii	residual annotator-marked identifier	person-identifying but outside the eight types

well below every public specialized baseline (0.557–0.972). Coverage and false positives must be read together, since a detector can raise CD by flagging nearly every document, as Presidio and the OpenMed models do.

Inference policy for long documents. Table 8 decodes the same PII-Tracer checkpoint under three long-input policies. Relative to the single window, sliding windows raise recall on documents at or above 10k characters from 0.687 to 0.975 and CD-m from 0.794 to 0.954, while character precision drops from 0.507 to 0.493; chunked decoding lies between the two.

E Prompts

The four LLM prompts of this work are shown below.

Gold Labeler Prompt (§3.3)

Task. You are an expert PII labeler producing gold labels for ... conversations. Your labels will train a downstream PII-masking model, so they must reflect industry-standard definitions of Personally Identifiable Information ...

Test. Would a knowledgeable reader, given this string and the surrounding context, be able to identify or contact a specific person? If the string instead identifies a publicly-listed organization, a public-figure persona, or a fictional vignette persona, the answer is no ...

Traps. Even when the string looks exactly like a name, address, phone, URL, or date, emit nothing if the context matches a trap: a named

Table 7: Entity-level results on the test split: consistent detection (CD), multi-mention CD (CD-m), cross-turn CD (xCD), and PII-free false-positive rate FP_0 (lower is better). Best per column in **bold**.

Detector	CD	CD-m	xCD	FP_0
Presidio	0.649	0.642	0.648	0.972
OpenAI PF	0.386	0.304	0.296	0.557
GLiNER2-PII	0.448	0.268	0.292	0.862
GLiNER-PII	0.472	0.349	0.334	0.901
Piirinha	0.128	0.090	0.097	0.685
OpenMed 44M	0.724	0.745	0.761	0.936
OpenMed 434M	0.709	0.743	0.759	0.935
GPT-5.4	0.545	0.279	0.275	0.321
GPT-5.6-sol	0.675	0.570	0.551	0.231
Claude Opus 4.8	0.530	0.239	0.222	0.386
Claude Sonnet 5	0.563	0.314	0.305	0.276
PII-Tracer	0.853	0.794	0.776	0.385

Table 8: PII-Tracer under three long-input decoding policies: a single window (trunc), non-overlapping windows (chunk), and 50%-overlap sliding windows (slide). $R_{<10k}$ and $R_{\geq 10k}$ split character recall at 10k characters.

Policy	P	R	F1	CD	CD-m	$R_{<10k}$	$R_{\geq 10k}$
trunc	0.507	0.830	0.629	0.853	0.794	0.956	0.687
chunk	0.502	0.944	0.656	0.916	0.894	0.956	0.931
slide	0.493	0.965	0.653	0.954	0.954	0.957	0.975

officer or staff member in an organizational role, reserved or documentation values, citation or tool tokens ...

Output. A single JSON object with two fields: a conversation-level sensitivity head and the token-level PII span list of `{label, text}` objects ...

Paraphrase Prompt (Figure 2, step 3)

Task. Rewrite a snippet from a real user–assistant chat so it is not verbatim the original, while preserving its meaning, intent, tone, language, and Markdown structure (headings, lists, tables, code fences ...).

Rule 1: placeholders. The text may contain special placeholder characters; each stands for a redacted value. Reproduce every placeholder exactly once, in the natural grammatical posi-

tion where its value belongs; never delete, duplicate, translate ...

Rule 2: de-identify context. Any real-world specific that is not a placeholder (company, product, or brand names; unusual place names; distinctive verbatim phrases ...) must become a generic or plausibly invented equivalent.

Output. Roughly the same length and the same language as the input; only the rewritten snippet, no preamble ...

Second-LLM Audit Prompt (§3.5)

Task. Audit synthetic-conversation text for a PII benchmark; all names, contacts, and IDs are supposed to be fabricated ...

Report. Only strings that look like a real, specific individual’s genuine identifying data that fabrication might have missed (an intact real email, phone, or handle; a real public figure named as the actual user ...).

Ignore. Generic names, places, and dates; clearly synthetic values.

Output. JSON `{"suspect": bool, "items": [{"text", "why"} ...]}`.

Frontier-LLM Detector Prompt (§4)

Task. You are a precise PII detector operating on text from a multi-turn assistant conversation. Extract every span that is personal information identifying a private individual ...

Rule. Mark a span only when, in this conversational context, it helps identify a private person; do not mark public figures, organizations, fictional or placeholder values ... A date counts only when it identifies a person ...

Output. A JSON list of `{text, type}` objects over the nine identifier types, with `text` copied verbatim from the input ...